Reading lamp: UK sourcing research (Farnell UK first, RS UK as fallback)

Researched 2 Nov 2026. No orders were placed and nothing was logged into.

How to read the "Verified?" column

Tag	Meaning
C	Order code verified: it appeared in a distributor product URL (`/dp/<code>` on Farnell, `/web/p/.../<code>` on RS, `/dp/SCxxxxx` on CPC).
P*	Price or stock comes from the search engine's cached copy of the distributor page. Farnell and RS block direct fetches (403/503), so these figures are undated, sometimes contradict each other, and are indicative only. Confirm on the live page.
DK	Live DigiKey UK fetch on 2 Nov 2026, used to sanity-check price and lifecycle. These are DigiKey prices, not Farnell's.
DS	Read from the manufacturer datasheet.
U	Unverified.

Farnell prices are GBP **ex VAT**. "@5+" means the price break at 5 units and above (minimum order is often 5 or 10 on cut tape). "—" means not seen.

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1. I2C constant-current LED driver (≥11 ch, ≥30 mA, TSSOP/HTSSOP)

Distributor	Order code	MPN	Mfr	Package	Price qty1 / qty10	Stock / lead	Lifecycle	Source URL
Farnell	**not found**	LP5012PWR	TI	TSSOP (DK lists "24-TSSOP"; check pin count against TI datasheet)	DK £1.18 / £0.864	DK 1,781	Active (DK)	Not in Farnell or RS search index (3 querie
Farnell	**not found**	LP5009PWR	TI	TSSOP	DK £0.81 / £0.587	DK 5,900	Active (DK)	—
Farnell	**2759859**	PCA9955BTWJ	NXP	HTSSOP-28 (exposed pad)	P* £2.68 / £2.28	P* 14,550, but the same cached page also shows "No Longer Manufactured" (2022-era text)	DK: Active, 28,575 in stock, £1.99 / £1.477	https://uk.farnell.com/nxp/pca9955btwj/le driver-constant-current-htssop/dp/275985
Farnell	2890538	PCA9955BTW/Q900J	NXP	HTSSOP-28	DK £2.64 (single price)	—	U	https://uk.farnell.com/nxp/pca9955btw-q900j/fm-i2c-bus-led-driver-16-ch-htssop/dp/2890538
Farnell	**3119169**	TLC59116IPWR	TI	TSSOP-28 (no pad)	P* £1.89 / £1.41	P* 1,988 in stock, 16-wk standard lead	DK: Active, 3,701, £2.00 / £1.484	https://uk.farnell.com/texas-instruments/tlc59116ipwr/ic-led-driver-8-b tssop-28/dp/3119169
RS	709-9161	TLC59116IPWR	TI	TSSOP-28	—	—	U	https://uk.rs-online.com/web/p/display-drivers/7099161
Farnell	2759860	PCA9956BTWY	NXP	HTSSOP-38	P* £2.69 qty1	P* 4,942 (another cache says out of stock / no longer stocked)	U	https://uk.farnell.com/nxp/pca9956btwy/le driver-constant-current-htssop/dp/275986

Notes:

- LP5012: 12 ch, 35 mA max, 2.7–5.5 V (DK). It is the only candidate here that runs from 2.7 V. PCA9955B is 3–5.5 V (Farnell listing); TLC59116 is 3–5.5 V. Both are fine on a Li-ion rail above 3.0 V.
- The PCA9955B's exposed pad needs hot air or a via-fed thermal pad for hand soldering. The TLC59116IPWR is plain TSSOP-28 at 5–120 mA per channel.
- The RS code 252-0675 from the brief is a TI LM3445MM/NOPB (triac-dimmable driver). It is unrelated, and RS no longer stocks it.

2. AVR MCU with UPDI, 16–32 KB, SOIC-20

Distributor	Order code	MPN	Mfr	Package	Price qty1 / qty10	Stock / lead	Lifecycle	Source URL	V
Farnell	**not found**	ATtiny1616-SNR	Microchip	SOIC-20W	DK £0.87 / —	DK 0; 1,413 due 11-Nov-2026; 20-wk lead	Active (DK)	—	I
Farnell	**2920834**	ATTINY1616-SFR	Microchip	SOIC-20 (125 °C grade)	P* £0.781–£0.936 @100+ (two conflicting caches)	stock not shown; page has a "restricted purchase" notice for some accounts	U	(code seen in Farnell result text; URL slug not captured)	(l t
Farnell	3630321	ATTINY1616-SF	Microchip	SOIC-20	—	—	U	https://uk.farnell.com/microchip/attiny1616-sf/microcontrollers-mcu-8-bit/dp/3630321	(
Farnell	**3018703**	ATTINY3216-SN	Microchip	SOIC-20	P* £1.04 qty1 (£0.862 @250+)	P* 647 in stock	DK £0.95 qty1	https://uk.farnell.com/microchip/attiny3216-sn/mcu-8bit-20mhz-soic-20/dp/3018703	(
Farnell	3630440	ATTINY3216-SNR	Microchip	SOIC-20W	P* £0.849–£1.02, **full reel only (1,500 MOQ)**	P* out of stock, 8-wk lead	U	https://uk.farnell.com/microchip/attiny3216-snr/microcontrollers-mcu-8-bit/dp/3630440	(
Farnell	2911436	ATTINY3216-SFR	Microchip	SOIC-20	—	—	U	https://uk.farnell.com/microchip/attiny3216-sfr/mcu-8bit-avr-20mhz-soic-20/dp/2911436	(
Farnell	3018697	ATTINY1614-SSN	Microchip	SOIC-14	—	—	U	https://uk.farnell.com/microchip/attiny1614-ssn/mcu-8bit-20mhz-soic-14/dp/3018697	(
Farnell	not found	AVR16DD20 / AVR32DD20 (SOIC-20)	Microchip	—	—	—	—	—	r
Farnell	3872794	ATTINY3226-SU (tinyAVR 2, SOIC-20)	Microchip	SOIC-20	—	—	U	https://uk.farnell.com/microchip/attiny3226-su/mcu-8bit-20mhz-soic/dp/3872794	(c r t k

CPC does not list ATtiny1616 or ATtiny3216 (search found only ATtiny85 and PICs).

3. Li-ion linear charger 4.2 V, SOT-23-5

Distributor	Order code	MPN	Mfr	Package	Price qty1 / qty10	Stock / lead	Lifecycle	Source URL
Farnell	**1332158** (cut tape), 1332158RL (re-reel)	MCP73831T-2ACI/OT	Microchip	SOT-23-5	P* £0.59 qty1, £0.581 @25	P* 8,094 in stock + 48,000 incoming. One cache mentions a 100 MOQ; an older (~2019) cache says NLM	DK: Active, 29,784, £0.57 / £0.57, 32-wk lead	https://uk.farnell.com/microchip/mcp73831-2aci-ot/li-ion-li-poly-charge-controller/dp/1332158
Farnell	1834890	MCP73831T-2ATI/OT	Microchip	SOT-23-5	—	—	U	https://uk.farnell.com/microchip/mcp73831-2ati-ot/batt-charger-li-ion-li-poly-5sot23/dp/1834890
CPC	SC18584	MCP73831T-2ATI/OT	Microchip	SOT-23-5	P* ~£0.51 qty1, ~£0.48 @25+ (the same cache also says "pricing unavailable")	—	U	https://cpc.farnell.com/microchip/mcp73832ati-ot/batt-charger-li-ion-li-poly-5sot23/dp/SC18584

Both suffixes are 4.2 V / 500 mA. They differ in the preconditioning and termination options, so check the datasheet's option table before swapping one for the other.

4. I2C temperature sensor

Distributor	Order code	MPN	Mfr	Package	Price qty1 / qty10	Stock / lead	Lifecycle	Source URL
Farnell	**not found**	TMP1075DR	TI	SOIC-8	DK £0.46 / £0.378	DK 0; 2,500 due 21-Dec-2026	Active (DK)	—
Farnell	not found	TMP1075DGKR	TI	VSSOP-8	DK £0.46	—	U	—
Farnell	**2841138** (cut tape); 2841138RL; full reel 4384267	LM75BD,118	NXP	SOIC-8	P* £0.527 @5+, £0.412 @50+	P* 47,843 in stock	Active per Farnell; DK £0.64 qty1	https://uk.farnell.com/nxp/lm75bd-118/temperature-sensor-open-drain/dp/2841138
Farnell	1854077	LM75BD,112	NXP	SOIC-8	—	—	**No Longer Manufactured**	https://uk.farnell.com/nxp/lm75bd/ic-temp-sensor-digital-8soic/dp/1854077
Farnell	2080523	MCP9808-E/MS	Microchip	MSOP-8	—	—	U	https://uk.farnell.com/microchip/mcp9808-e-ms/temp-sensor-i2c-20-to-100-deg/dp/2080523
Farnell	3124240	TMP102AIDRLR	TI	SOT-563 (0.5 mm pitch, awkward to hand-solder)	—	—	U	https://uk.farnell.com/texas-instruments/tmp102aidrlr/temperature-sensor-1deg-c-sot/dp/3124240
Farnell	8455970	TMP75AIDR	TI	SOIC-8 (LM75 footprint)	—	—	U	https://uk.farnell.com/texas-instruments/tmp75aidr/ic-temperature-sensor-smd-soic8/dp/8455970
Farnell	2396743	PCT2075D,118	NXP	SOIC-8 (LM75 drop-in, ±1 °C)	—	—	U	https://uk.farnell.com/nxp/pct2075d-118/temperature-sensor-2deg-cel-soic/dp/2396743

5. P-MOSFET, SOT-23, ≥|20| V, Rds(on) ≤ 60 mΩ @ −2.5 V, |Vgs(th)| ≤ 1 V

Distributor	Order code	MPN	Mfr	Package	Price qty1 / qty10	Stock / lead	Lifecycle	Source URL	Verified?
Farnell	**2543533**	DMG2305UX-7	Diodes	SOT-23	P* £0.197 @5+, £0.141 @50+	P* 346,272 (cache dated around Oct 2024; the same cache also shows a "no longer stocked" notice)	DK: Active, **0 stock**, backorder, £0.28 / £0.173	https://uk.farnell.com/diodes-inc/dmg2305ux-7/mosfet-p-ch-20v-sot-23-3/dp/2543533	C, P* (conflicting) DK, DS
Farnell	**1843688**	DMG3415U-7	Diodes	SOT-23	—	—	Datasheet header: **NOT RECOMMENDED FOR NEW DESIGN, use DMP2045U**. DK: 51,820 in stock, £0.39 / £0.238	https://uk.farnell.com/diodes-inc/dmg3415u-7/mosfet-p-ch-20v-4a-sot-23/dp/1843688	C, DK, DS
Farnell	not checked	DMP2045U-7	Diodes	SOT-23	DK £0.33 / £0.202	DK 54,628	Active (DK)	—	DK (45 mΩ max @ −4.5 V, Vgs(th) max −1 V; the −2.5 V figure is unverified)
Farnell	not checked	PMV48XP,215	Nexperia	SOT-23	DK £0.45 qty1	DK unclear	Active (DK)	—	DK: 55 mΩ @ −4.5 V, **Vgs(th) max −1.25 V, so it fails the ≤1 V criterion**

Distributor	Order code	MPN	Mfr	Package	Price qty1 / qty10	Stock / lead	Lifecycle	Source URL	Verified?
Farnell	1470106	SI2323DS-T1-E3	Vishay	SOT-23	—	—	**Obsolete** (DK)	https://uk.farnell.com/vishay-siliconix/si2323ds-t1-e3/mosfet-p-20v-4-7a-sot-23/dp/1470106	C. Avoid
Farnell	1858668	DMP2305U-7	Diodes	SOT-23	—	—	U (listings disagree: 45 vs 60 mΩ)	https://uk.farnell.com/diodes-inc/dmp2305u-7/mosfet-p-ch-20v-4-2a-sot-23/dp/1858668	C
—	not checked	DMP2035U	Diodes	—	—	—	—	—	not checked (search budget ran out)

Datasheet figures (DS):

- **DMG2305UX**: Rds(on) 52 mΩ typ / **100 mΩ max** @ −2.5 V; 40 / 52 mΩ @ −4.5 V; Vgs(th) −0.5 to −0.9 V. It meets the 60 mΩ target only on the typical value.
- **DMG3415U**: Rds(on) 40 typ / **53 mΩ max** @ −2.5 V; 42.5 mΩ max @ −4.5 V; Vgs(th) −0.3 to −1.0 V. It meets the spec on maximum values, but it is NRND.

6. Schottky ≥1 A, ≥20 V, low Vf

Distributor	Order code	MPN	Mfr	Package	Price qty1 / qty10	Stock / lead	Lifecycle	Source URL
Farnell	1510673	PMEG2010AEH,115 (no plain "PMEG2010EH" found)	Nexperia	SOD-123F	P* £0.459 @5+, £0.284 @10+	P* **28 in stock**	DK: Active, 0 stock	https://uk.farnell.com/nexperia/pmeg2010ae115/diode-schottky-1a-20v-sod-123f/dp/1510673
Farnell	1510684	PMEG3010EH,115	Nexperia	SOD-123F	P* £0.352 @5	P* out of stock, **54-wk lead**	DK: Active, 0 stock	https://uk.farnell.com/nexperia/pmeg3010eh115/diode-schottky-1a-30v-sod-123f/dp/1510684
Farnell	1510687	PMEG3020EH,115	Nexperia	SOD-123F	P* £0.219 @100–499	—	U	https://uk.farnell.com/nexperia/pmeg3020eh115/diode-schottky-2a-30v-sod-123f/dp/1510687
Farnell	**1431044**	MBR120VLSFT1G	onsemi	SOD-123FL	P* £0.354 @5+, £0.208 @50+ (minimum 5)	P* 6,731 in stock	DK: Active, 108,104, £0.46 / £0.279; Vf 340 mV max @ 1 A	https://uk.farnell.com/on-semiconductor/mbr120vlsft1g/diode-schottk1a-20v-sod-123fl/dp/1431044
Farnell	1858617	B130LAW-7-F	Diodes	SOD-123	P* ~£0.398 (2022 cache)	P* 39,250 (old)	U	https://uk.farnell.com/diodes-inc/b130law-7-f/diode-rec-schot-1a-30v-sod123/dp/185861

7. USB-C receptacle (power-only 6-pin preferred)

Distributor	Order code	MPN	Mfr	Package	Price qty1 / qty10	Stock / lead	Lifecycle	Source URL	Verifiec
Farnell	**3648642** (cut tape); 3648648 = full reel, 1,000 MOQ	USB4125-GF-A	GCT	6-pos USB-C, R/A, SMT + THT shell, 3 A / 48 V	P* £0.593 qty1 (cut tape), £0.402 @200+	P* 39,682	DK £0.46 qty1 (status not shown)	https://uk.farnell.com/gct-global-connector-technology/usb4125-gf-a/usb-conn-2-0-type-c-r-a-rcpt-6pos/dp/3648642	C, P*, DI
Farnell	3777661	USB4125-GF-A-0190	GCT	variant of the above	—	—	U	https://uk.farnell.com/global-connector-technology/usb4125-gf-a-0190/usb-conn-type-c-r-a-rcpt-6pos/dp/3777661	C
Farnell	3973076 (also seen: 3946920)	USB4135-GF-A	GCT	6-pos, horizontal	—	—	U	https://uk.farnell.com/gct-global-connector-technology/usb4135-gf-a/usb-conn-type-c-r-a-rcpt-6pos/dp/3973076	C
Farnell	2924867	USB4085-GF-A	GCT	16-pos THT, 5 A (orientation listed inconsistently)	P* £1.40 qty1	P* 54	U	https://uk.farnell.com/gct-global-connector-technology/usb4085-gf-a/usb-conn-2-0-type-c-r-a-rcpt-16pos/dp/2924867	C, P*

Distributor	Order code	MPN	Mfr	Package	Price qty1 / qty10	Stock / lead	Lifecycle	Source URL	Verified
Farnell	3367461	12401610E4#2A	Amphenol	24-pos USB 3.1 SMD (fine pitch, poor for hand soldering)	P* ~£0.86	—	U	https://uk.farnell.com/amphenol-fci/12401610e4-2a/usb-conn-3-1-type-c-rcpt-smd/dp/3367461	C
—	not checked	Würth 632723300011, Molex 2171750001	—	—	—	—	—	—	not checked

8. TVS / ESD for 5 V VBUS

Distributor	Order code	MPN	Mfr	Package	Price qty1 / qty10	Stock / lead	Lifecycle	Source URL
Farnell	**2691565** (also 2691664)	SMF5.0A	Littelfuse	SOD-123FL; uni, 5 V Vrwm, 9.2 V Vc, 200 W	P* £0.148 @5+ (minimum 5)	P* ~13–14k	DK £0.37 (single price)	https://uk.farnell.com/littelfuse/smf5-0a/tvs-diode-200w-5v-unidir-sod-123fl/dp/2691565
Farnell	3361273	SZSMF5.0AT1G	Littelfuse	SOD-123FL (AEC-Q101)	—	—	U	https://uk.farnell.com/littelfuse/szsmf5-0at1g/tvs-diode-200w-5v-unidir-sod-123fl/dp/3361273
Farnell	1757865	PESD5V0S1UL,315	Nexperia	**SOD-882 (1.0 × 0.6 mm DFN), not SOD-323**	—	—	U	https://uk.farnell.com/nexperia/pesd5v0s1ul-315/diode-esd-protection-uni-sod-882/dp/1757865
—	not checked	ESD9B5.0ST5G, Vishay SMF5V0A	—	—	—	—	—	—

9. Tactile switch, SMD, top-actuated, 2.5–4.3 mm tall (3 per board)

Distributor	Order code	MPN	Mfr	Package	Price qty1 / qty10	Stock / lead	Lifecycle	Source URL	Verified
Farnell	3780598	PTS526 SM15 SMTR2 LFS (Farnell lists "SMTR2"; plain "SMTR" not found)	C&K	5.2 × 5.2 mm, 160 gf, **1.5 mm actuator height (DK), below the 2.5 mm minimum**	DK £0.12	DK 250,785	Active (DK)	(code from Farnell listing text)	C (text), DK
Farnell	**3941596** (also 2320087, an older listing at £0.242 @10+)	PTS645SM43SMTR92 LFS	C&K	6 × 6 mm, 4.3 mm high, 160 gf	P* £0.192 @10+ (**minimum 10**)	P* 8,960	DK £0.27 (status not shown)	https://uk.farnell.com/c-k-components/pts645sm43smtr92-lfs/tactile-switch-0-05a-12vdc-160gf/dp/3941596	C, P*, DK
—	not checked	TL3301AF160QG, B3FS-1000P, Würth 434121025816	—	—	—	—	—	—	not checked

10. Slide switch SPDT, right-angle SMD

Distributor	Order code	MPN	Mfr	Package	Price qty1 / qty10	Stock / lead	Lifecycle	Source URL	Verified?
Farnell	**9575146** (9575146RL re-reel, £3.50 reel charge)	PCM12SMTR	C&K	SMD, horizontal / side-actuated, 300 mA @ 6 V DC	P* £0.85 / £0.72	P* 26,658	DK £0.96 (status not shown)	https://uk.farnell.com/c-k-components/pcm12smtr/switch-spdt-0-3a-6vdc-smd/dp/9575146	C, P*, DK
—	not checked	JS102011SAQN, Würth 450301014042	—	—	—	—	—	—	not checked

11. JST PH 2-pin battery header

Distributor	Order code	MPN	Mfr	Package	Price qty1 / qty10	Stock / lead	Lifecycle	Source URL	Verified?
Farnell	**9492615**	S2B-PH-SM4-TB(LF)(SN)	JST	SMD right-angle (side entry)	P* £0.478 @10+ (**minimum 10**)	P* 219,429	U	https://uk.farnell.com/jst-japan-solderless-terminals/s2b-ph-sm4-tb-lf-sn/connector-header-smt-r-a-2mm-2way/dp/9492615	C, P*
Farnell	9491856	B2B-PH-K-S(LF)(SN)	JST	THT top-entry	—	—	U	https://uk.farnell.com/jst-japan-solderless-terminals/b2b-ph-k-s-lf-sn/header-top-entry-2way-2mm/dp/9491856	C
CPC	CN14073	B2B-PH-K-S(LF)(SN)	JST	THT top-entry	—	—	U	https://cpc.farnell.com/jst-japan-solderless-terminals/b2b-ph-k-s-lf-sn/header-top-entry-2way-2mm/dp/CN14073	C
CPC	CN14084	S2B-PH-K-S(LF)(SN)	JST	THT side-entry	—	—	U	https://cpc.farnell.com/jst-japan-solderless-terminals/s2b-ph-k-s-lf-sn/header-side-entry-2way/dp/CN14084	C

12. JST SH 4-pin (Qwiic / STEMMA QT) header and cable

Distributor	Order code	MPN	Mfr	Package	Price qty1 / qty10	Stock / lead	Lifecycle	Source URL	Verified?
Farnell	**1830839**	SM04B-SRSS-TB(LF)(SN)	JST	SMD R/A 1 mm, 4-way	P* £0.492 @10+, £0.408 @100+ (category page)	P* 54,298	U	https://uk.farnell.com/jst-japan-solderless-terminals/sm04b-srss-tb-lf-sn/connector-header-smt-r-a-1mm-4way/dp/1830839	C, P*
RS	515-1513	SM04B-SRSS-TB(LF)(SN)	JST	same	—	—	U	https://uk.rs-online.com/web/p/pcb-headers/5151513	C
RS	**288-4665**	SparkFun KIT-15081 Qwiic Cable Kit (10 cables, 50–500 mm, includes a breadboard-jumper style)	SparkFun	cable kit	P* £11.09 ex VAT (another cache: £9.63)	—	U	https://uk.rs-online.com/web/p/development-tool-accessories/2884665	C, P*
Farnell	**1679110**	SHR-04V-S-B (4-way SH housing)	JST	housing	P* £0.0487 @100+ (**minimum 100, so £4.87**)	P* 362,052	U	https://uk.farnell.com/jst-japan-solderless-terminals/shr-04v-s-b/connector-housing-4way/dp/1679110	C, P*
Farnell	**1679137**	SH3-SH3-28150 (pre-crimped lead, socket to socket, 150 mm, red, 28–32 AWG)	JST	lead	P* £0.303 @10+ (**minimum 10**)	P* 8,391	U	https://uk.farnell.com/jst-japan-solderless-terminals/sh3-sh3-28150/lead-socket-socket-32-28awg-150mm/dp/1679137	C, P*
Farnell	**1679139**	SH3-SS5-28150 (socket to **free end**, 150 mm, red)	JST	lead	—	—	U	https://uk.farnell.com/jst-japan-solderless-terminals/sh3-ss5-28150/lead-socket-free-end-150mm-red/dp/1679139	C
Farnell	1679138	SH 300 mm socket-to-socket lead, red (MPN not captured)	JST	lead	—	—	U	(code from Farnell result text)	C (text)

No single-cable Adafruit 4210/4401/5384 or SparkFun PRT Qwiic cables turned up at Farnell, RS or CPC. Two ways to get a cable from these distributors:

- **Pre-made:** the RS SparkFun kit above.

- **Built from Farnell parts:** 4 × SH3-SH3-28150 plus 2 × SHR-04V-S-B makes a 150 mm cable. 4 × SH3-SS5-28150 plus 1 × SHR-04V-S-B makes a pigtail with one bare end. Every lead is red, so mark the wires yourself. No crimp tool is needed.

13. White LEDs, 2835/3030, CRI ≥ 90, 3 V class, If max ≥ 60 mA, ~2700 K and ~5700–6500 K (8 of each)

Distributor	Order code	MPN	Mfr	CCT / CRI	If typ / max, Vf	Flux, angle	Price	Stock	Lifecycle	Source URL
Farnell	**3498849**	BXEN-27S-11M-3C-00-0-3 (Thrive, SMD 2835, 3.5 × 2.8 × 0.7–0.8 mm)	Bridgelux	2700 K / **97 typ**	150 mA test = max; 3.1 V typ	51 lm @ 150 mA (45 lm @ 85 °C); 116°	P* £0.10 @5+ … £0.053 @1000+	P* 370	DK: Active (DK has no stock)	https://uk.farnell.com 27s-11m-3c-00-0-3/c white-110lm-w-2700
Farnell	3498851	BXEN-40S-11M-3C-00-0-3	Bridgelux	4000 K / 97	150 mA; 3.1 V	56 lm; 116°	—	—	U	https://uk.farnell.com 40s-11m-3c-00-0-3/c white-120lm-w/dp/34
Farnell	**3498853**	BXEN-50S-11M-3C-00-0-3	Bridgelux	**5000 K** / 97	150 mA; 3.1 V	59 lm; 116°	P* £0.175 @5+, £0.116 @100+	P* 13,220–37,747	U	https://uk.farnell.com 50s-11m-3c-00-0-3/c white-127lm-w-5000
Farnell	not found	BXEN-65S-11M-3C-00-0-3 (6500 K Thrive)	Bridgelux	6500 K / 97	150 mA	57 lm	DK reel-only (170k)	DK 0	Active (DK)	—
Farnell	3227684	GW QSLM31.QM-H1HV-XX57-1 (OSCONIQ S 3030, quantum dot)	ams OSRAM	3000 K / **≥90 min**	65 mA test, 180 mA max; 2.75 V	43.5 lm; 120°	—	P* 2,132 (Aug-2023 cache)	U	https://uk.farnell.com semiconductors/gw-q h1hv-xx57-1/led-hb-w 5lm-3000k/dp/32276
Farnell	2470235 / 2470239	158563427 (2700 K) / 158563460 (6000 K), WL-SWTP **5630**	Würth	**CRI 85 typ (DK), below 90**	100 mA test, 300 mA max; 3.0 V	34 / 39 lm; 120°	P* £0.402 qty1 (6000 K); DK £0.40	P* 4,700 (6000 K)	Active (DK)	https://uk.farnell.com elektronik/15856346 white-39lm-6000k/dp
Farnell	4691153	GW JTLPS1.SM-H5H8-XX38-1-G (DURIS E 2835)	ams OSRAM	2700 K / CRI not stated	65 mA; 2.71 V	120°	—	—	U	https://uk.farnell.com group/gw-jtlps1-sm-h g/led-warm-white-27 120deg/dp/4691153
RS	225-0696	GW QTLTS2.QM-GWH5-XX58-1-65-R33 (OSCONIQ E 2835)	ams OSRAM	2700 K / ≥90	2.71–2.9 V	120°	—	**unavailable, being discontinued**	NRND/EOL	https://uk.rs-online.com/web/p/le
RS	264-8164	JB2835BWT-G-H40GA0000-N0000001	Cree LED	4000 K? / U	—	—	—	—	**Discontinued**	https://uk.rs-online.com/web/p/le
RS	232-0718	L130-2780HA30000B1 (LUXEON 3030 HE Plus)	Lumileds	2700 K / **80**	240 mA max; 2.71 V	30–33.5 lm	—	—	U	https://uk.rs-online.com/web/p/le
RS	137-2002 etc.	NF2W757GT-F1-5080, NF2L757GT-F1-3080/-4080	Nichia	Ra80 versions only seen	—	—	—	several marked unavailable	U	https://uk.rs-online.com/web/p/le
—	not found	Cree JB2835AWT 90 CRI, Samsung LM281B+ / LM302Z+, LUXEON 2835 90 CRI, DURIS S5 PSLR31 white	—	—	—	—	—	—	Farnell's Cree J 2835 parts are flagged NLM. DigiKey's Samsung LM281B listings are mostly Obsolete or reel-only	—

14. RGB LED (separate dies preferred, ≥30 mA per die)

Distributor	Order code	MPN	Mfr	Package / config	Die If; λd (nm)	Price	Stock	Lifecycle	Source URL
Farnell	not found	CLV6A-FKB	Cree	—	—	—	—	—	—
Farnell	2414841	CLP6C-FKB-CKNPRGJBB7A363	Cree LED	PLCC-6 (separate dies)	20 mA class per listing (U); 624 / 540 / 480	—	—	U	https://uk.farnell.com cknprgjbb7a363/led- 6/dp/2414841
Farnell	3217313	CLX6F-FKC-CKNNQDGBB7A363	Cree LED	PLCC-6	15 / 10 / 10 mA test; 624 / 540 / 480	—	**no longer stocked**	NLS	https://uk.farnell.com cknnqdgbb7a363/ful reel/dp/3217313
Farnell	2981675	LRTB GVTG (no LRTB GVSR found)	ams OSRAM	MULTILED	20 mA	—	**no longer stocked**	NLS	https://uk.farnell.com semiconductors/lrtb- 15cd-630mcd-smd/d
Farnell	1827013	LRTBGFUG	OSRAM	6-lead MULTILED, 2.8 mm (separate dies)	R 40 / G 50 / B 50 mA	—	—	U	https://uk.farnell.com black-body-rgb/dp/1
Farnell	**2322115** (2322115RL re-reel flagged NLM in one cache)	150141M173100	Würth	3528, **common anode** (DK)	30 mA each (Farnell), 20 mA test (DK); 625 / 520 / 470; 270 / 950 / 230 mcd	P* £0.53 **inc VAT** qty1 (~£0.44 ex), £0.38 inc @500+; DK £0.37	P* 4,731; DK 3,205	Active (DK)	https://uk.farnell.com elektronik/150141m 270mcd-950mcd-23
Farnell	2401105	ASMB-MTB0-0A3A2	Broadcom	PLCC-4	20 mA	—	—	**Discontinued / NLM**	https://uk.farnell.com 0a3a2/led-hb-rgb-0-(
Farnell	2368159	ASMT-YTB7-0AA02	Broadcom	PLCC-6	20 / 20 / 20 mA (below 30)	—	—	U	https://uk.farnell.com 0aa02/led-plcc6-rgb- mcd/dp/2368159
Farnell	1863244	ASMT-YTC2-0AA02	Broadcom	PLCC-6	R 50 / G 30 / B 30 mA	P* £1.44 / £0.958	—	**No Longer Manufactured**	https://uk.farnell.com 0aa02/led-plcc6-rgb- body/dp/1863244
Farnell	1554799	ASMT-QTB0-0AA02	Broadcom	PLCC-4	R 50 / G 30 / B 30 mA	—	—	U	https://uk.farnell.com 0aa02/led-plcc4-rgb- surface/dp/1554799
Farnell	**2335791**	KAAF-5050RGBS-13 (no "AAAF5051" found)	Kingbright	5050, clear lens (pad config: check datasheet; believed 6-pad separate dies)	R 50 / G 30 / B 30 mA; 625 / 525 / 470; 1.4 cd / 1.4 cd / 420 mcd; Vf 2.5 / 3.3 / 3.5 V	P* £1.70 @5+ (minimum 5)	P* 5,189	U	https://uk.farnell.com 5050rgbs-13/led-sm 420/dp/2335791
Farnell	2112155	703-1039	Multicomp	5050 RGB	30 / 30 / 30 mA	—	—	U	https://uk.farnell.com 1039/led-rgb-5050/d

15. Indicator LED and passives (format examples)

Part	Distributor	Order code	MPN	Notes	Source URL	Verified?
Red LED 0603	Farnell	3796306	MP008277 (Multicomp Pro)	626 nm, 220 mcd @ 20 mA, 2.4 V	https://uk.farnell.com/multicomp-pro/mp008277/led-red-220mcd-626nm-0603/dp/3796306	C
Red LED 0603	Farnell	1716774	OVS-0608 (Multicomp Pro, also "703-0069")	625 nm, 300 mcd, 2.1 V	https://uk.farnell.com/multicomp/ovs-0608/led-0603-super-bright-red/dp/1716774	C
10 k 1 % 0603	Farnell	2447230 (cut tape) / 2447230RL / 2446547 (reel)	MCWR06X1002FTL (Multicomp Pro)	MPN pattern `MCWR06X<value code>FTL`, e.g. 100 R = MCWR06X1000FTL, code 2447227	https://uk.farnell.com/multicomp-pro/mcwr06x1002ftl/res-10k-1-0-1w-0603-thick-film/dp/2447230	C
10 k 1 % 0603	Farnell	9238603	RC0603FR-0710KL (Yageo)	100 k = RC0603FR-07100KL, code 9238727	https://uk.farnell.com/yageo/rc0603fr-0710kl/res-10k-1-0-1w-0603-thick-film/dp/9238603	C

Part	Distributor	Order code	MPN	Notes	Source URL	Verified?
1 µF 10 V X5R 0603	Farnell	3013385	CL10A105KP8NNNC (Samsung)	—	https://uk.farnell.com/samsung-electro-mechanics/cl10a105kp8nnnc/cap-1uf-10v-mlcc-0603/dp/3013385	C
4.7 µF 10 V X5R 0603	Farnell	3013398	CL10A475KP8NNNC (Samsung)	—	https://uk.farnell.com/samsung-electro-mechanics/cl10a475kp8nnnc/cap-4-7uf-10v-mlcc-0603/dp/3013398	C
10 µF 10 V X5R 0603	Farnell	3013389	CL10A106MP8NNNC (Samsung, ±20 %)	The ±10 % "KP8" version was not seen with a code	https://uk.farnell.com/samsung-electro-mechanics/cl10a106mp8nnnc/cap-10uf-10v-mlcc-0603/dp/3013389	C
10 µF 10 V X5R 0603	Farnell	3764144	0603X106K100CT (Multicomp Pro, ±10 %)	—	https://uk.farnell.com/multicomp-pro/0603x106k100ct/cap-10uf-10v-mlcc-0603/dp/3764144	C
100 nF 50 V X7R 0603	Farnell	1759122	MC0603B104K500CT (Multicomp Pro)	P* ~642k in stock	https://uk.farnell.com/multicomp/mcca000255/mlcc-0603-x7r-50v-100nf/dp/1759122RL	C, P*
100 nF 50 V X7R 0603	Farnell	1362556	CC0603KRX7R9BB104 (Yageo)	—	https://uk.farnell.com/yageo-phycomp/cc0603krx7r9bb104/cap-mlcc-x7r-100nf-50v-0603/dp/1362556	C

Farnell order codes are 7 digits. Re-reel versions add "RL", and a full reel usually has its own separate code. Multiples and minimum quantities vary by line (often 5, 10, 50 or 100) and were not checked line by line.

16. Programming adapter (SerialUPDI or native UPDI)

Distributor	Order code	Product	Price	Source URL	Verified?
Farnell	1329311	FTDI TTL-232R-3V3 (FT232R, 3.3 V I/O, 6-pin SIP, 1.8 m)	P* £19.41 or £25.85 (two caches)	https://uk.farnell.com/ftdi/ttl-232r-3v3/cable-usb-to-ttl-level-serial/dp/1329311	C, P*
Farnell	1740365	FTDI TTL-232R-3V3-WE (wire-ended)	—	https://uk.farnell.com/ftdi/ttl-232r-3v3-we/cable-usb-ttl-3-3v-wire-end/dp/1740365	C
CPC	SC10142	FTDI TTL-232R-3V3	—	https://cpc.farnell.com/ftdi/ttl-232r-3v3/cable-usb-to-ttl-level-seri-converter/dp/SC10142	C
Farnell	2915518	Microchip PG164100 MPLAB Snap (supports AVR UPDI through MPLAB X/IPE)	P* ~£8.24 (looks unusually low; confirm)	https://uk.farnell.com/microchip/pg164100/in-circuit-debugger-1-2v-to-5/dp/2915518	C, P*
Farnell	4216621	Microchip PG164150 MPLAB PICkit 5	P* ~£88.83	https://uk.farnell.com/microchip/pg164150/in-circuit-debugger-2-7v-to-5v/dp/4216621	C, P*
—	not checked	Adafruit 954 / 5335	—	—	not checked (search budget ran out)

Buying from Farnell UK as an individual, and CPC

- **Farnell UK online** has no minimum order quantity or value. Standard delivery is **free at £40 and over**, **£8.99 below £40**, and Express costs £11.99. Items shipped from US (Newark) stock add a flat **£15.95 per order**; an older terms page says £9.99. The terms also reserve a "small order handling charge" on orders below £40 ex VAT. Prices are ex VAT, so a personal buyer adds 20 %. Source: uk.farnell.com/help-delivery-information and the terms-of-purchase page, via search snippets; the live page returned 503.
- **Whether Farnell UK sells to individuals is unverified.** No page in this research said it does or doesn't. Its registration and trade-account pages mention card orders and sole traders.

- **CPC (cpc.farnell.com)** is Farnell's consumer and trade arm. Its registration asks "Business or Personal", so individuals are accepted. Free standard delivery applies on UK mainland orders **over £24.99 ex VAT** (some pages say £20), and Click & Collect is free with no minimum.
- Parts confirmed at CPC: MCP73831T-2ATI/OT SC18584 (~£0.51 ex VAT, against ~£0.59 at Farnell for the -2ACI), FTDI TTL-232R-3V3 SC10142, JST B2B-PH-K-S CN14073, JST S2B-PH-K-S CN14084, and the SparkFun/Pimoroni maker range. Not at CPC: ATtiny1616/3216, the SMD JST S2B-PH-SM4-TB, and most of the specialised LEDs and ICs. CPC prices were checked for only one part, so "cheaper at CPC" is unverified apart from the MCP73831.

Method and limits

- Farnell, RS and Octopart pages return 403/503 to fetch tools, and no bot-protection workaround was attempted. Farnell and RS data comes from domain-restricted web search. Order codes are reliable because they are in the URLs; prices and stock are cached snippets.
- RS prices were not obtained for any part.
- The web-search budget ran out near the end. These were **not checked**: TMP1075 at RS; Farnell codes for DMP2045U-7, PMV48XP and DMP2035U; ESD9B5.0ST5G and SMF5V0A; TL3301AF160QG, B3FS-1000P and Würth 434121025816; JS102011SAQN and Würth 450301014042; Würth 632723300011 and Molex 2171750001; OSRAM LRTB G6TG; Samsung LM302Z+; Adafruit 954/5335; CPC prices beyond the MCP73831.

Recommended pick per function

1. **LED driver:** Farnell **2759859 PCA9955BTWJ** (HTSSOP-28 with exposed pad, 16 ch, 57 mA). It is the first candidate from the list that Farnell carries; LP5012 is not at Farnell or RS. Confirm the page isn't flagged "no longer stocked" (DigiKey shows it Active with 28k in stock). If it is flagged, use **3119169 TLC59116IPWR** (plain TSSOP-28, ~£1.89, 1,988 in stock). Both need VCC ≥ 3.0 V.
2. **MCU:** Farnell **2920834 ATtiny1616-SFR** if the live page shows stock in small quantities. Otherwise **3018703 ATtiny3216-SN** (SOIC-20, ~£1.04, 647 in stock, same pinout, 32 KB).
3. **Charger:** Farnell **1332158 MCP73831T-2ACI/OT** (~£0.59, ~8k in stock).
4. **Temperature sensor:** Farnell **2841138 LM75BD,118** (SOIC-8, ~£0.53 @5+, ~48k in stock). TMP1075 is not at Farnell.
5. **P-MOSFET:** Farnell **2543533 DMG2305UX-7** is active, but its datasheet Rds(on) at −2.5 V is 52 mΩ typ and **100 mΩ max**. **1843688 DMG3415U-7** meets 53 mΩ max at −2.5 V and Vgs(th) ≤ 1 V, but it is **NRND** (successor DMP2045U, Farnell code not checked). Matt decides whether 60 mΩ is a typ or max limit.
6. **Schottky:** Farnell **1431044 MBR120VLSFT1G** (SOD-123FL, 340 mV max at 1 A, ~£0.35 @5+, 6.7k in stock). The Nexperia PMEG parts are out of stock or on a 54-week lead.
7. **USB-C:** Farnell **3648642 GCT USB4125-GF-A** (6-pin power-only, ~£0.59, ~40k in stock).
8. **TVS:** Farnell **2691565 Littelfuse SMF5.0A** (SOD-123FL, ~£0.15 @5+). PESD5V0S1UL is SOD-882, which is not hand-solderable.
9. **Tactile switch:** Farnell **3941596 C&K PTS645SM43SMTR92 LFS** (6 × 6 × 4.3 mm, minimum 10 at ~£0.19). The PTS526 SM15 is only 1.5 mm tall.
10. **Slide switch:** Farnell **9575146 C&K PCM12SMTR** (~£0.85 / £0.72 @10, 26k in stock).
11. **Battery connector:** Farnell **9492615 JST S2B-PH-SM4-TB(LF)(SN)** (minimum 10 at ~£0.48). For a through-hole part, use **9491856 B2B-PH-K-S**.
12. **Qwiic header and cable:** header Farnell **1830839 SM04B-SRSS-TB(LF)(SN)**. For the cable, either RS **288-4665 SparkFun KIT-15081** (~£11, 10 cables) or build one from Farnell **1679110** SHR-04V-S-B housings with **1679137** (both ends) or **1679139** (one bare end) 150 mm SH leads.
13. **White LEDs:** warm, Farnell **3498849 Bridgelux BXEN-27S-11M-3C-00-0-3** (2700 K, CRI 97, 3.1 V, 150 mA, 51 lm, 116°, ~£0.10; only 370 in stock). Cool, **3498853 BXEN-50S-11M-3C-00-0-3**, but it is **5000 K**, not 5700–6500 K, so this is Matt's call. The 6500 K BXEN-65S is not stocked in the UK, and no CRI ≥ 90 part at 5700–6500 K was found at Farnell or RS.
14. **RGB LED:** separate dies at ≥30 mA, Farnell **2335791 Kingbright KAAF-5050RGBS-13** (R 50 / G 30 / B 30 mA, ~£1.70; confirm the pads are independent in the datasheet). Cheaper but common anode: **2322115 Würth 150141M173100** (3528, 30 mA, ~£0.44 ex VAT).
15. **Passives:** red LED **3796306** MP008277. Resistors: Multicomp Pro MCWR06X…FTL (e.g. **2447230** = 10 k) or Yageo RC0603FR-07… (**9238603**). Capacitors: Samsung CL10A105KP8NNNC **3013385**, CL10A475KP8NNNC **3013398**, CL10A106MP8NNNC **3013389**, and 100 nF MC0603B104K500CT **1759122**.
16. **Programmer:** Farnell **2915518 MPLAB Snap PG164100** (native UPDI, confirm the ~£8 price), or **1329311** (or CPC **SC10142**) FTDI TTL-232R-3V3 for SerialUPDI.